When OXXO Moves Into the Block: An Econometric Analysis of the Impact of OXXO on Criminal Activity in Bogotá

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https://github.com/sophiearisti/thesis_economics.git

1 Abstract

2 Acknowledgements

3 Introduction

Although Bogotá, the capital city of Colombia, has experienced a significant reduction in crime since the 1990s (J. F. Godoy et al., 2018; Rodríguez González, 2017), delinquency and insecurity remain major public concerns. The implementation of preventive programs and policies by Bogotá's administrations since the 2000s, along with other measures targeting crime hotspots (J. F. Godoy et al., 2018), has not effectively deterred the high levels of personal theft and the growing perception of insecurity that have affected the daily lives of citizens in recent years (Cámara de Comercio de Bogotá, 2026). This continued prevalence suggests that additional urban dynamics may shape patterns of criminal activity in Bogotá, and that alternative approaches and other possible factors to understanding crime in the city may still need to be explored.

One factor discussed in urban crime literature is the role of commercial establishments in shaping local patterns of criminal activity (Askey et al., 2018; Davis et al., 2021; Dugato et al., 2026; Nazzari et al., 2026; Wellford et al., 1997). Within this category of establishments, OXXO, a convenience store chain that has steadily expanded across Bogotá since 2009, can be a case study for studying this phenomenon in the city (M. C. P. Godoy, 2025-03-17). In particular, due to their long operating hours, strategic locations in high-traffic areas and commercial corridors (Marcos, 2022), availability of basic goods, and constant customer flows throughout the day and night, the arrival of this chain may alter surrounding patterns of mobility and social interaction in ways that could influence nearby criminal activity (Dugato et al., 2026; Marcos, 2022).

In fact, there is a duality in the crime literature regarding the relationship between convenience stores and crime. On the one hand, these establishments have been identified as *crime generators* because they increase foot traffic and create incidental criminal opportunities (Dugato et al., 2026). Similarly, Wellford et al. (1997) classify them as robbery-prone commercial establishments due to the availability of cash on hand. On the other hand, Davis et al. (2021) offer a different perspective, arguing that convenience stores may also function as crime deterrents by increasing natural surveillance and generating "eyes on the street" through extended operating hours and improved lighting. These two lines of reasoning highlight the importance of analysing the role of convenience stores within the surrounding criminal environment, particularly because, as Dugato et al. (2026) argue, addressing the urban conditions that facilitate criminal behavior may be more effective for long-term crime reduction than traditional repressive approaches.

OXXO's footprint in Colombia is centralized, with over 300 of the more than 500 stores in the country concentrated in Bogotá (M. C. P. Godoy, 2025-03-17). This high density within a single metropolitan area provides an"urban laboratory" to conduct a granular, interurban crime analysis. By exploiting the staggered arrival of these establishments across Bogotá's Zonal Planning Units (UPZs), this thesis evaluates the causal impact of OXXO openings on local criminal activity. Specifically, I examine how these effects vary by crime category, and whether the impact intensifies or dissipates over time following the store's inauguration.

To estimate this impact, I constructed a quarterly panel dataset covering the period from 2015 to 2022, aggregated at the level of the UPZs, Bogotá's sub-district administrative divisions. The panel combines Colombia's public georeferenced crime records from the National Police's Statistical, Criminal, Contraventional, and Operational Information System (SIEDCO) with a geolocated database of OXXO establishments that accounts for both opening and, when applicable, closure dates. Using this information, I calculated quarterly crime rates for each UPZ and identified the number of OXXO stores operating within each unit over time.

Additionally, I employed Two-Way Fixed Effects (TWFE) and the Callaway and Sant'Anna (C&S) estimator, along with their corresponding event-study specifications, as the main empirical strategies of this investigation. The TWFE model was useful for estimating the average relationship between the expansion of OXXO stores and crime rates across UPZs over time. However, because staggered treatment adoption may generate bias under treatment-effect heterogeneity, the C&S estimator was also implemented to obtain more robust estimates and assess the consistency of the TWFE results. Finally, event-study analyses were conducted to examine the dynamic evolution of treatment effects over time and to evaluate the parallel trends assumption for both empirical approaches.

At the end, I found that the staggered entry of OXXO stores had no statistically significant effect on criminal activity across Bogotá. Using both aggregated specifications through the TWFE model and dynamic specifications through event-study analyses, I found no evidence that the expansion of OXXO affected different types of crime in the city. Additionally, after controlling for the presence of other similar commercial establishments located near OXXO stores, such as hard-discount retailers, which have also been associated with changes in criminal activity in Bogotá, the estimated effects remained statistically insignificant. These findings suggest that, both in the short and long run, OXXO stores do not appear to significantly alter Bogotá's security dynamics. This may be because ANOTHER EXPLANATION.

Despite extensive research on urban crime in Bogotá, no study has systematically examined the causal impact of convenience store proliferation, particularly OXXO, on local crime patterns. Existing investigations have focused on public transport infrastructure (Bacares, 2013; Bacares et al., 2013a; García et al., 2005), large-scale police interventions (Blattman et al., 2017, 2021), alcohol sales restrictions (Mello et al., 2013), and socioeconomic determinants (Escobar, 2012). One recent study examined the relationship between Tiendas D1 hard-discount stores and crime (D. López et al., 2024), but OXXO represents a distinct retail model characterized by 24-hour operations and a higher density of "grab-and-go" items, which may create different criminal opportunities compared to the limited operating hours of hard-discount formats.

In the next section I briefly introduce OXXO's expansion across Bogotá, and the city's crime rates over the years. Then, in Section 3 I present the related literature about the spatial nexus of crime and convenience stores. In Sections 4 and 5 I describe the data sources used and the general descriptive statistics. After that I explain in Section 6 the econometric models used and in Section 7 I show the results obtained. In Section 8 I include some robustness checks related with the unit of analysis used, the control and treatment groups, and the specification of some variables. Finally, Section 9 discusses possible mechanisms that may influence the results obtained.

4 Background

4.1 The expansion of OXXO in Bogotá

Although OXXO has been operating in Colombia for almost 17 years, it only consolidated itself as one of the main convenience store chains in the country after the pandemic (El Economista, 2026). In fact, in Bogota the number of establishments rose

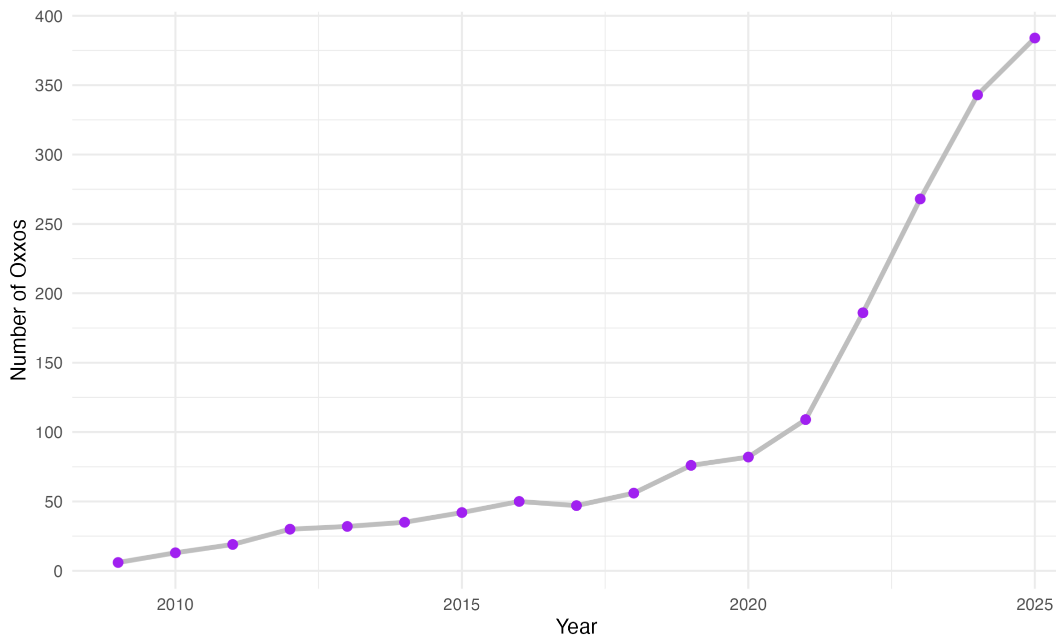


Figure 1: **Yearly evolution of the number of OXXOs in Bogotá**

sharply from 82 in 2020 to 384 in 2025, this represents an increment of over 368% in only five years. Figure 1 depicts this accelerated growth in Bogotá, as it shows the number of OXXO locations in the capital by year since its arrival. This OXXO expansion can be due to some of its intrinsic characteristics; for example, 95% of its establishments function in a 24-hour format (OXXO Colombia, 2026), providing a competitive advantage by being the primary option of consumers whenever they need to satisfy spontaneous consumption needs (M. C. López, 2026).

The proliferation of OXXO stores began in the northern part of the city before spreading throughout the eastern region. Figure 2 illustrates this evolution, showing that OXXO is now present in nearly 70% of the city's Zonal Planning Units (UPZ) , these being units that group neighborhoods based on socioeconomic stratification, land use, housing characteristics, and the living conditions of their inhabitants (Alcaldía Mayor de Bogotá, 2021). Furthermore, it can be observed that OXXO presence has a primary concentration in the northeastern neighborhoods of Bogotá. Currently, the only areas lacking OXXO establishments are in the far southern reaches of the city. These places correspond to UPZs with lower levels of economic performance, socioeconomic stratification, and urban transport connectivity (Bocarejo & Tafur, 2013;

Guzman & Bocarejo, 2017; PRISM, 2025), where residents continue to rely heavily on traditional neighborhood stores (Rico-Pencue et al., 2021).

This expansion across the years positively correlates with the inherent characteristics of northeastern Bogotá, where high traffic, spatial accessibility, and high-income residential clusters provide an ideal environment for the convenience store model (citation). Figure 3 illustrates this behavior by mapping the city's arterial roads, main avenues, and Transmilenio stations (Bogotá's Bus Rapid Transit system); it demonstrates that by 2025, OXXO locations tended to cluster along these major transport corridors. Furthermore, Section 7 provides a statistical analysis confirming the positive correlation between these variables and store locations.

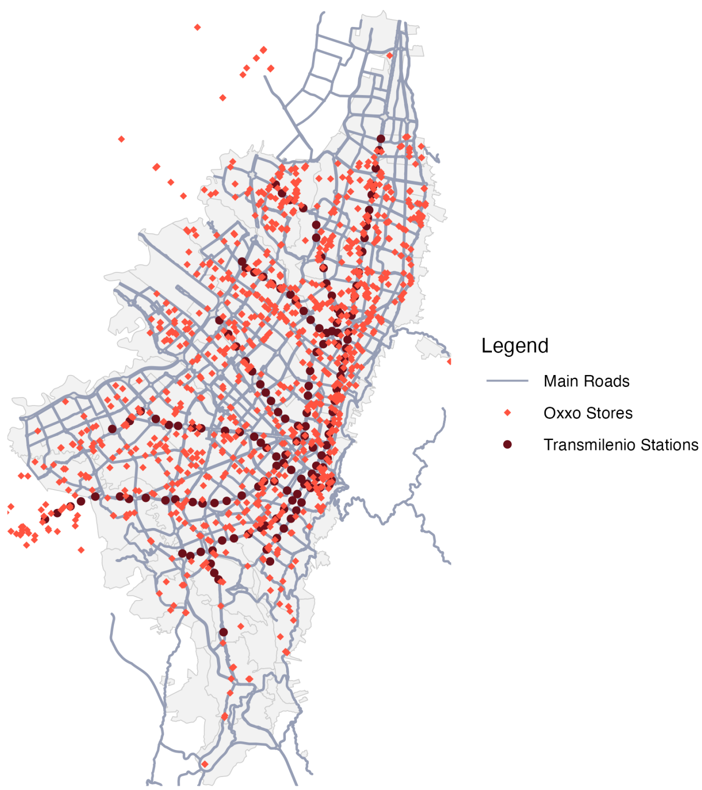


Figure 3: **Bogota's OXXO stores, Transmilenio stations and main roads in 2025**

OXXO locations are centrally administered by FEMSA with the purpose of standardizing processes, logistics, services, and products (OXXO Colombia, 2026). This

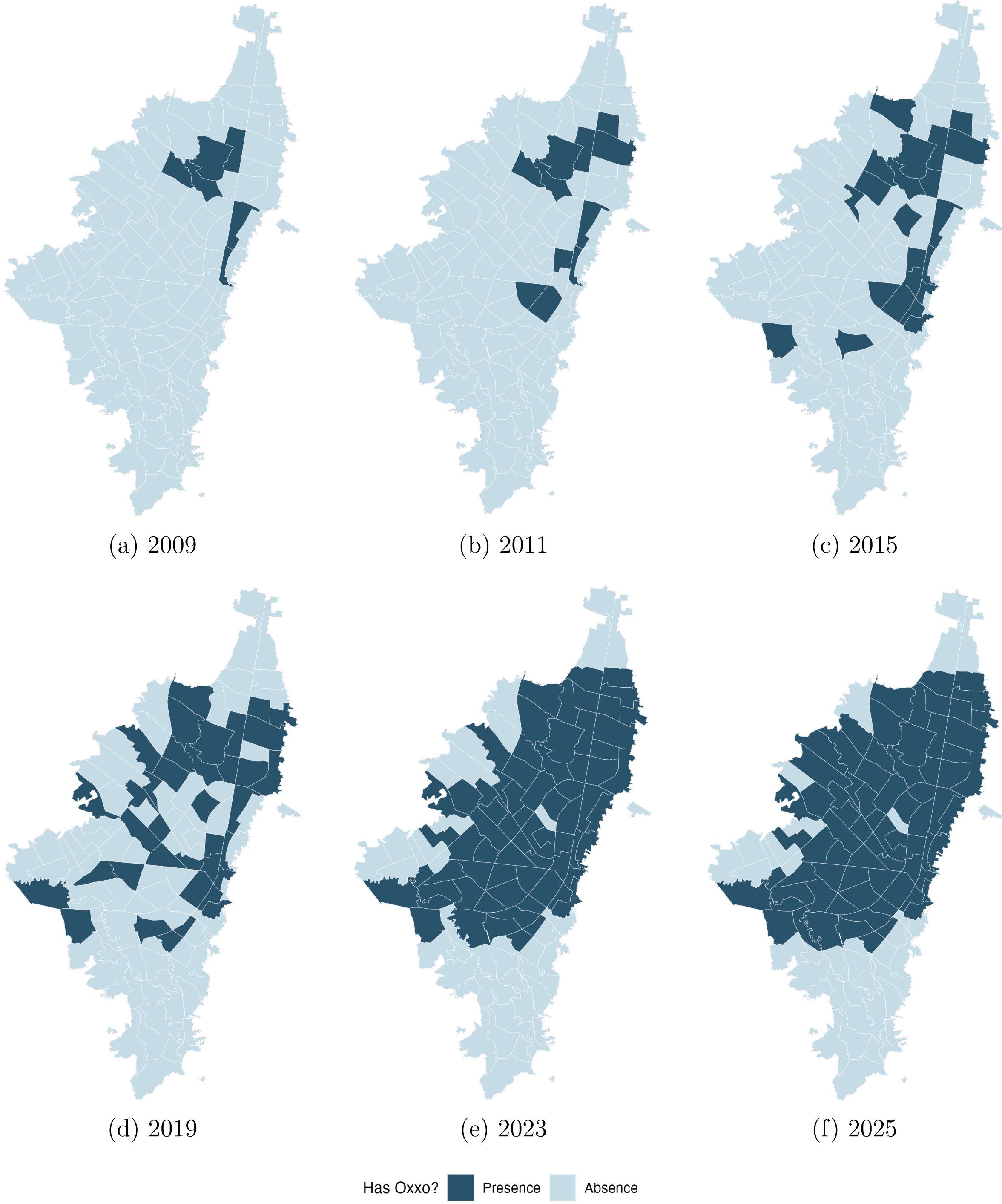


Figure 2: **Evolution of the presence of OXXO by UPZ**
Note: This map only indicates which UPZs have OXXOs, it does not consider the number of locations present in each Unit

implies that store placement is strategically determined according to the company's broader commercial objectives rather than being randomly distributed across space. Such non-aleatory allocation may complicate econometric identification, since the characteristics that attract OXXO establishments may also affect crime behavior.

The concern about endogenous OXXO placement, however, may be mitigated through the use of TWFE estimators, because many determinants of store location are either time-invariant or evolve only gradually over the sample period, allowing UPZ fixed effects to absorb a substantial portion of the selection process. Nonetheless, some bias may persist if OXXO entry responds to time-varying local trends correlated with the outcome. For example, FEMSA may prioritize expansion into UPZs experiencing increases in commercial activity, population density, household income, pedestrian flows, or improvements in public security, all of which could independently influence the dependent variable. In such cases, TWFE estimates could partly capture these underlying trends rather than the causal effect of OXXO entry alone.

4.2 Criminality in Bogotá

During the 1980s and 1990s, Bogotá was among the most unsafe cities in Colombia, experiencing a surge in narcoterrorism and the assassination of political leaders; in fact, in 1991, it reached a rate of 1,287 crimes per 100,000 inhabitants, exceeding that of Medellín (1,160) and Cali (537) (Rodríguez González, 2017). However, at the end of the 90s and beginning of the 2000s, crime started to decrease; for example, between 1993 and 2002 Bogotá saw an estimated 65% decrease in the homicide rate, while robbery fell by about 33%. This shift was driven by citizen culture programs, large-scale urban renewal projects, and institutional reforms implemented by the majors of this period (Mockus, 2012; Sánchez et al., 2003). In recent years, however, urban crime has shown a fluctuating behavior. Following the 2019 covid pandemic, criminality had a rebound, with personal theft surpassing pre-pandemic levels in 2022 and 2023 (Policía Nacional de Colombia, 2026). By 2024, despite continued institutional efforts and a decrease in crime rates in the city (Policía Nacional de Colombia, 2026), 69.3% of households believed insecurity had raised (Cámara de Comercio de Bogotá, 2025).

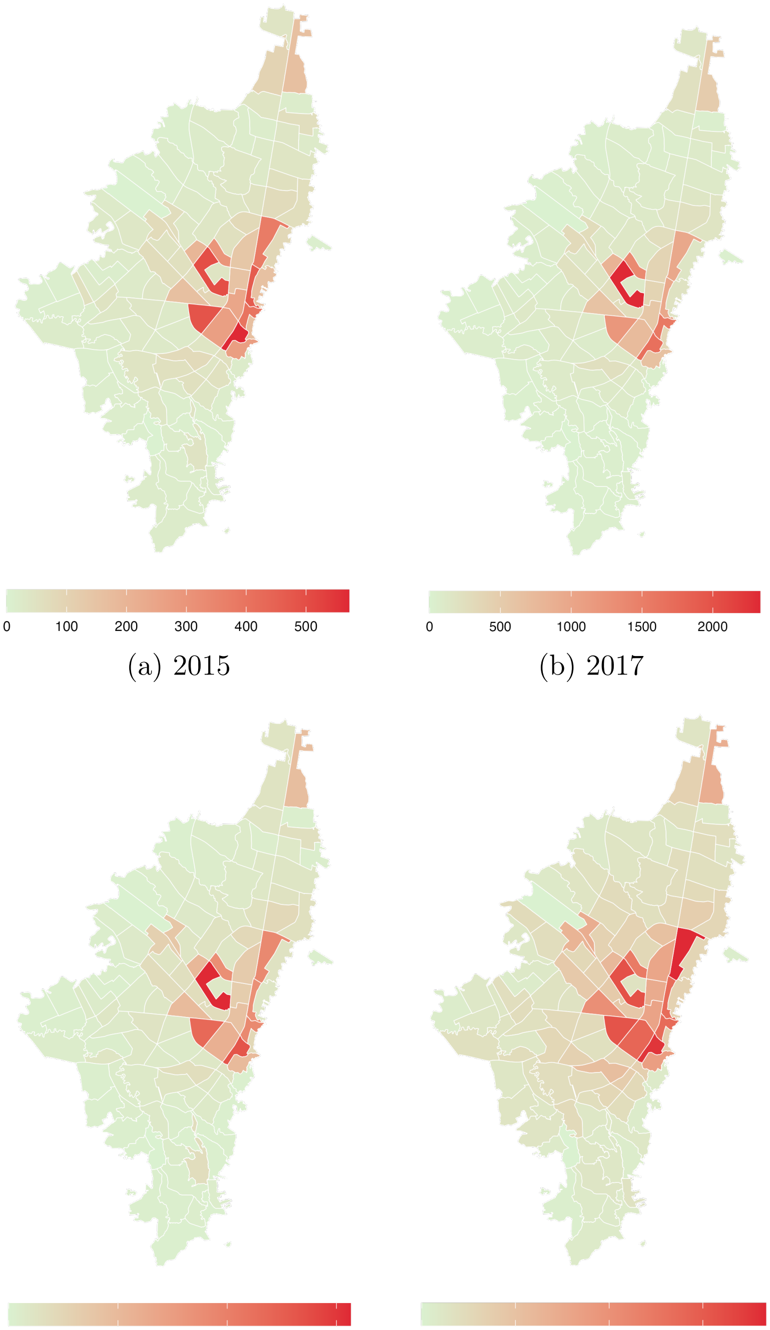
Currently, crime in Bogotá, as in most cities, tends to cluster in specific zones (Enriquez et al., 2014; Giménez-Santana et al., 2018). The most insecure Localities (a unit of territorial division that gathers many UPZ) are primarily located in the southeastern region of the city, a pattern associated with the presence of criminal organizations, high levels of informality, and deficiencies in human development (ProBogotá Región, 2024). However, this spatial concentration does not imply that all categories of crime peak in the same areas. While homicide rates generally follow this pattern, personal theft exhibits a distinct geographic distribution, with higher concentrations in the northeastern sector of the city (ProBogotá Región, 2023).

Figure 4 illustrates this behaviour as it displays how personal theft rates distribute throughout Bogotá´s UPZs over the years (Appendix D shows how other crimes distribute). These maps also highlight that between 2015 and 2021 (and even later, as it can be noted in ProBogota's annual security reports) this tendency has not drastically changed. This persistence suggests that there are environmental characteristics that keep favoring robberies or attracting offenders. Among these characteristics are high pedestrian traffic, a dense concentration of commercial land use, and proximity to the city's main transit corridors.

It is precisely within these high-activity environments that OXXO has focused its expansion. Consequently, for this research, factors that favor crime clustering must be taken into account. If OXXO stores tend to locate in inherently high-crime areas, it becomes essential to assess whether their presence contributes to an increase in crime or whether both the stores and offenders are are simply drawn to the same high-activity 'nodes.' In this sense, OXXO locations may function either as crime generators or as crime attractors.

5 Related Literature

In the context of urban economics, spatial econometrics and crime literature, various studies have explored how built environment features and infrastructure shape crime activity in cities. A number of empirical analyses have investigated this by exploring the causal impact of urban public transit on crime. Xu et al., 2021, for example, assessed the impact of two new subway lines implemented in 2017 in ZG city, China, on thefts in the surrounding areas. By employing a Difference-in-Differences (DiD) framework, they found that in the short run, theft increased by 15%. Others, on the other hand, have



(a) 2015 (b) 2017

(c) 2019 (d) 2021

Figure 4: **Evolution of theft rates by UPZ**

Note: The theft rate per 10,000 inhabitants for each unit was calculated using rate smoothing techniques (Empirical Bayes smoothing). This approach aimed to reduce the influence of outlier UPZs that had remarkably high rates due to their small populations rather than high theft activity. Still,after this transformation, I had to exclude two UPZs and set their rates to zero, as they continued to show extreme values that overshadowed the crime activity of other areas. In fact, one UPZ is an airport, and the other is mainly rural and contains many storage facilities instead of dwellings.

ventured to explore the paper of new urban policies and infrastructure interventions on this topic. Domínguez and Asahi, 2022 paper is one case, as they study the causal effect of ambient light on crime in Santiago, Chile, by using the "Daylight Saving Time" policy as a natural experiment. Due to this policy, they could find that an increase of one hour of ambient light exposure between 7 and 9 reduced thefts by 30%. These studies emphasize that urban interventions do not just shift crime spatially, but actively reshape the rational choice of offenders by creating or eliminating environmental opportunities for illicit activity.

Retail chains can be considered as built environment features that may influence urban crime activity. While different studies have extensively explored their role, these previous inquiries have their own limitations. A substantial body of research dates back to the pre-2000s era, meaning that their findings do not contemplate the current urban dynamics of modern cities. Nonetheless, these foundational studies provide a critical perspective on the underlying mechanics of crime. For instance, early investigations focused on identifying the characteristics that influenced the propensity of convenience stores to be targeted for robbery. They identified that stores that operated for longer hours at night were more likely to be robbed, as there are fewer people around (D'Alessio & Stolzenberg, 1990). Others found that these types of stores were more prone to be robbed due to the lack of visible security such as police presence, security guards, or effective alarm systems (Dugato et al., 2026; Wellford et al., 1997). Although useful, they fail to account for convenience stores' role on crime of their immediate surroundings.

On the other hand, other recent publications have examined the influence of analogous commercial establishments on urban crime. While some of these studies do not focus exclusively on convenience stores, they utilize other retail outlets with similar operational profiles which can serve as proxies for what this thesis aims to investigate. Askey et al. (2018) is and example as the authors pooled data from fast-food restaurants and convenience stores in Seattle. They used sales revenue as a proxy for human activity to count the different levels of operational intensity across locations, and aimed to explain crime patterns in the immediate surroundings of these establishments. Their results indicate that stores with higher average sales were associated with with a 7.4% increase in expected crime counts. Conversely, Feng et al., 2019 contribute to this discussion by finding that alcohol outlets influenced the likelihood of aggravated assault and street robbery in New York City. Their findings reveal differential impacts depending on the type of outlet (if it is a bar, a grocery store, or other similar establishment); for instance, the authors highlighted that in Manhattan and the Bronx, proximity within two blocks of a grocery store increased the relative risk of street robbery by more than sixfold.

Another relevant investigation is the one of (Shin, 2024). Shin used a DiD approach to invetigate the impact of dollar stores on localized violent crime patterns in Chicago. To ensure the treatment and control groups were as homogeneous as possible, the author created localized areas using a 250-feet buffer for the treatment group and a 250-to-354-feet outer ring as the control. At the end, this research found that store openings lead to a 21.% increase in violent crime (0.17 incidents per quarter), a result primarily driven by a 38% surge in robberies.

Although these studies offer a useful baseline of how convenience stores, and specifically OXXO, impact nearby crime, they present an incomplete perspective as they exclusively explored this nexus in cities of high-income nations. To build a more robust theory, these hypotheses must be tested within Latin American cities which have particular urban dynamics. Many papers argue that convenience stores are normally located in socio-economically disadvantaged areas where poverty, social disorganization, and lack of community resources exacerbate crime risks (Dugato et al., 2026; Shin, 2024), but in cities like Bogotá the situation is not analogous. OXXO in Bogotá, for instance, does not follow this rule as it instead prioritizes to open locations in middle to high income neighborhoods, and places with high commercial activity and influx of people.

In Latin America, the economic literature regarding convenience stores remains limited. Recent investigations have explored their causal role on various labor market indicators (Delgado et al., 2024; Marcos, 2022). However, despite this growing interest in labor dynamics, the spatial relationship between these establishments and urban security in the region has been largely overlooked. Specifically in Bogotá crime literature has primarily focused on public transport, general socioeconomic urban caracteristics, and policy intervention impact on crime (Bacares et al., 2013b; Moreno García, 2005).

The investigations that have actually explored this topic are counted. For example, Jaramillo López et al., 2021 used the same method as (Shin, 2024) to study the effect of D1 hard discount stores on the crime around the areas where they open in Bogotá. Although D1s are not convenience stores per se, they attract a similar type of costumers and and often share similar spatial locations with OXXO outlets . By analyzing the period of 2016 to 2019 and employing the same information utilized in this thesis (within SIEDCO infromation system), the authors found that D1 openings caused a 33.4% average reduction in crime density per block. They attributed this impact to the 'eyes on the street' phenomenon, suggesting that the increased pedestrian flow acts as a form of informal guardianship that deters criminal activity (Jacobs, 1961; Jaramillo López et al., 2021).

As noted, the body of literature that explores the impact of convenience stores on crime in Latinamerica is reduced, and virtually non-existent in the Colombian context. In this thesis I aim to fill this gap by evaluating the impact of OXXO store openings on local crime in Bogotá. Given their rapid expansion and market dominance, OXXO stores serve as a primary actor and a representative proxy for modern convenience retail in the city. Additionally, this analisis is particularly relevant in a context where crime remains a public policy priority and a growing concern for Bogota's inhabitants.

6 Data used and unit of analysis

For this thesis I use UPZs as the unit of analysis. UPZs work as a feasible division of Bogotá as the city maintains continuity and consistency of data across the years, and security interventions are normally focalized by UPZ (cita. Also, these Units maintain urbanistic and socioeconomic homogeneity by grouping similar neighborhoods, and are the middle point between Localities and Street blocks (manzanas).

Meanwhile, the results of this study employ four main data sources (see appendix A). For measuring crime rate in Bogotá I used the georeferenced crime events stored in SIEDCO [1] spanning from 2015 to 2022. SIEDCO is Colombia's official Police database that consolidate real-time citizen reports and administrative records of criminal activity. Here, for each incident recorded the system saves details including type of crime (e.g., homicide, personal robbery, commercial burglary, or cell phone theft), the precise geographic coordinates of the event, date and time of the incident, and the victim's

[1] Access at https://oaiee.scj.gov.co/agc/rest/services/Tematicos_Pub/SIEDCO_Delitos_Pub/FeatureServer

gender. With this information, I aggregated the crime data at the UPZ-year level, categorizing incidents by offense type. To ensure comparability across years and Units, I normalized these counts using annual population estimates for each UPZ. This allowed for the construction of a crime rate per 10,000 inhabitants, as shown in this equation:

$$crime\_type\_index_{upz,year} = \frac{total\_incidents_{upz,year}}{population_{upz,year}}; \quad \forall upz, year \tag{1}$$

Following this normalization, the rates were smoothed using an Empirical Bayes technique to reduce the influence of extreme values in UPZs with small populations, where rates tend to be more unstable; for example, UPZs like El Mochuelo and El Dorado had this particulatiries.

The period from 2015 to 2022 is appropriate for this analysis, as it encompasses the years during which OXXO experienced its most significant expansion across Bogotá. This timeframe provides sufficient temporal and spatial variation, as a larger number of UPZs enter treatment during the study period rather than being treated prior to 2015. Prior to 2015, the lack of new store openings in other UPZs would result in an insufficient number of newly treated units, limiting the identifying variation necessary for using TWFE.

Additionally, using data from the Registro Único Empresarial y Social (RUES)[2], Colombia's central business registry, I identified the opening and closing years of each OXXO store. Furthermore, with Google Maps API I obtained the precise georeferenced location of each store. Concerns may arise regarding the accuracy of this matching process, particularly for stores that are no longer in operation. To address this, I manually verified each location using official OXXO sources, and as a result 95% of establishments were properly assigned.

I also did this same process with D1, Justo&Bueno and ARA hard discount chains. The reason why I also took into account these brands is because they often share similar spatial locations with OXXO outlets; in fact, retailers tend to cluster in certain areas to take advantage of agglomeration benefits (Konishi, 2005; Seong et al., 2022). Also, Given that prior research suggests these establishments may influence both crime and commercial activity (Delgado et al., 2024; Jaramillo López et al., 2021), their inclusion as control variables is important to evaluate the validity of the results.

[2]Access at https://www.rues.org.co

Finally, I used the Quality of Life Survey of Bogotá (2007)[3] to characterize the baseline socioeconomic conditions of treated and never-treated UPZs prior to the arrival of OXXO in 2009. Additionally, data from Mapas Bogotá were used to construct time-invariant socioeconomic and environmental controls at the UPZ level. These variables are included for descriptive purposes to assess the non-random spatial distribution of OXXO stores, although they are absorbed by the two-way fixed effects (TWFE) specification.

7 Descriptive Statistics

Before proceeding with the causal analysis, it is important to illustrate the preliminary differences between controlled and treated UPZs, not only in terms of crime activity, but also regarding socioeconomic and accessibility variables. Table 1 presents the staggered arrival of OXXO stores across UPZs per year between 2015 to 2022. Although the panel was constructed quarterly, the arrival of locations do not variate drastically among quarters per year. This table shows that there was an increase in treated UPZs starting in 2019, while a substantial number of control units remained available for comparison over the years.

This variation shown may not be as drastic as OXXO´s stores growth depicted in figure 1. This suggests that some UPZs host a disproportionately higher number of establishments in order to account for the overall increase in OXXO stores across Bogotá. Moreover, this raises the question of whether treatment intensity should be explicitly considered in the analysis. Although Figure 5 does not provide a detailed spatial distribution of OXXO stores at the UPZ level over time, it illustrates the evolution of the average number of stores per cohort. The figure shows that, in early periods, differences between cohorts were relatively small (less than one store on average), whereas in later periods these differences became more pronounced. This pattern suggests that variation in treatment intensity may play a role in shaping the estimated relationship between OXXO presence and crime, and therefore warrants further evaluation.

[3] Access at https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-2007-bogota

Table 1: Distribution of UPZs and OXXO stores over time

Year	**Not yet treated**	**Treated**	**Total OXXOs**
2015	90	20	38
2016	89	21	46
2017	87	23	46
2018	87	23	55
2019	81	29	74
2020	72	38	81
2021	70	40	108
2022	51	59	173
Total UPZs	**110**	–	–

Note: This table presents the annual count of UPZs by treatment status and the total number of OXXO stores in Bogotá. A UPZ is treated if it has at least one store.

On the other hand, Figure 5 presents preliminary evidence the behavior of different types of crime rates across the periods. Specifically, they report the difference in mean crime rates between treated and not-yet-treated UPZs for each quarter, along with an indication of whether these differences are statistically significant at the 5% level. For instance, Panel 6a shows that, for personal theft, there was a statistically significant difference between groups during the quarters from 2015-Q1 to 2019-Q1. This indicates that, on average, treated UPZs exhibited higher crime rates than those not yet treated.

These results are consistent with expectations, as theft is the crime category that is most likely to be positively associated with the presence of OXXO stores. This finding aligns with the hypothesis that such establishments may function as crime attractors by increasing the flow of suitable targets. However, this pattern may also reflect the possibility that OXXO stores are disproportionately located in UPZs with inherently higher theft rates. Moreover, these graphs indicate that differences for other types of crime are generally not statistically significant. This fits the pattern expected, as crimes such as sexual violence or vehicle and motorcycle theft are less likely to be influenced by OXXO presence. These offenses often rely on different spatial and social dynamics, such as secluded areas or long-term parking patterns, which remain largely unaffected by convenience retail expansion cita.

Table 2 also shows preliminary differences between treated and never treated groups

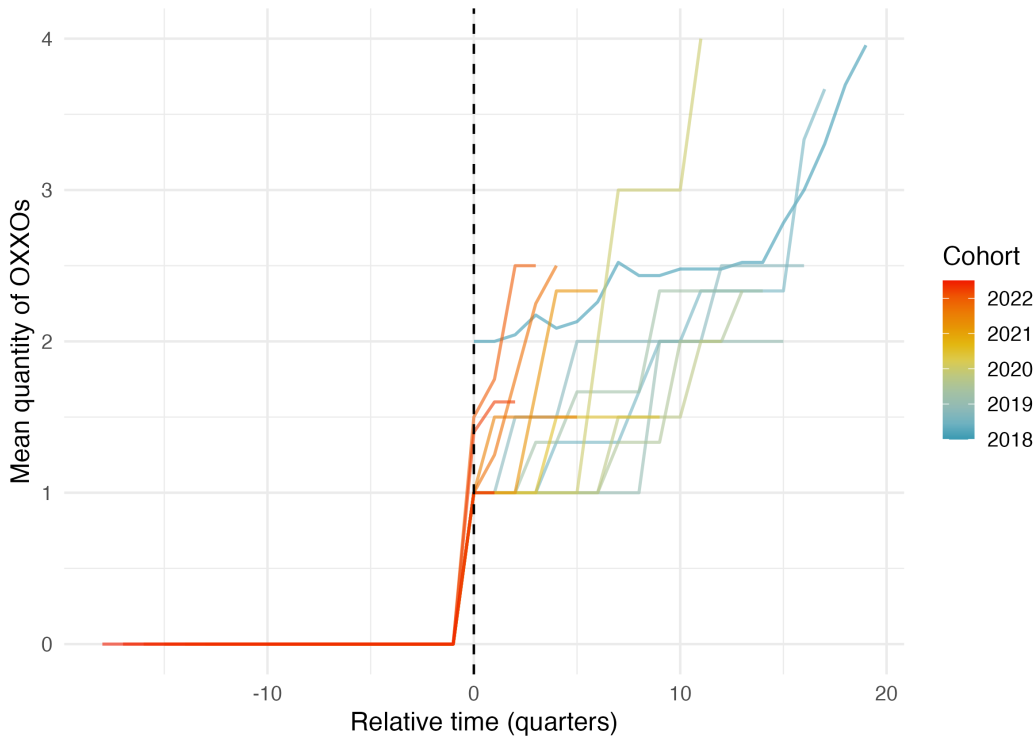
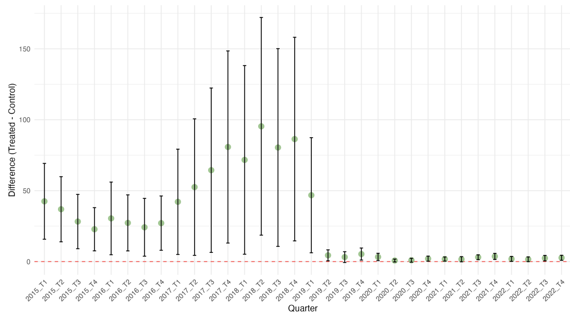
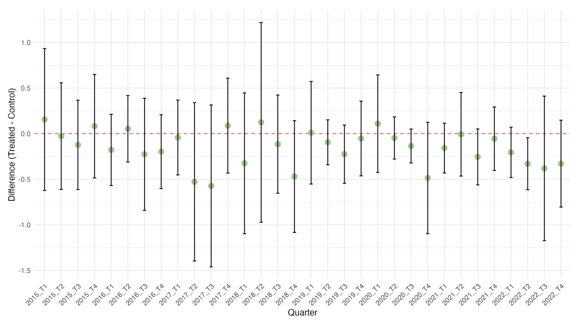


Figure 5: **Treatment Intensity: Evolution of OXXO Quantity by Cohort**

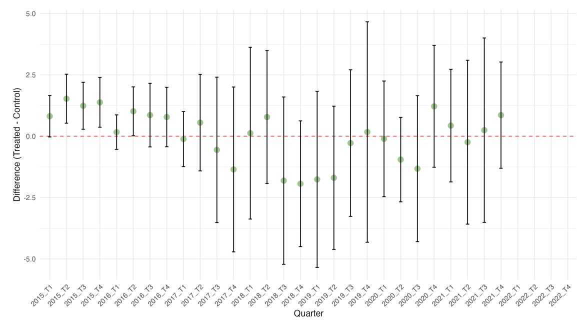
Note: The figure displays the average number of OXXO stores per UPZ, centered at the first quarter of entry ($t = 0$). Each line represents a different opening cohort (quarter-year). The color gradient indicates the chronological order of the cohorts, where darker lines represent [older/newer] cohorts. The dashed vertical line marks the start of the treatment.



(a) Theft rate



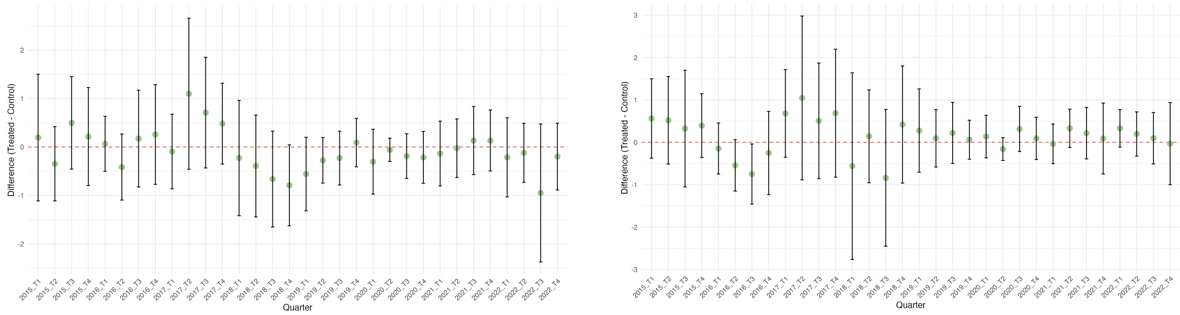
(b) Homicide rate



(c) Sexual Offenses rate

Figure 6: **Difference in Means Analysis (Part I)**

with fixed baseline controls. The variables chosen help to visualize that OXXOs are located in areas where there is higher traffic, accessibility, quality of life, and purchasing power. Moreover, these results indicate that potential market size may not be adequately captured by population measures alone (such as total inhabitants or household size in 2007), but rather by indicators of economic activity and consumption capacity. Since OXXO's product mix consists primarily of non-essential daily goods and convenience items rather than vital necessities, this retail chain prioritizes locations with high-income transit rather than just high residential density. These results are consistent with what was confirmed by Marcos (2022), where convenience store chains in Mexico were found to be distributed by similar characteristics.



(d) Theft to Motorbike rate (e) Theft to Vehicle rate

Figure 6: **Difference in Means Analysis (Part II)**

Note: This figure displays the difference-in-means tests for various crime categories across Bogota's UPZs. The confidence intervals represent 95% significance. If the intervals do not cross the zero line (red), the difference between treated and control units is statistically significant.

Table 2: Differences of means between treated and never treated for fixed controls

	Never treated	**Treated**	**Difference of means**
Baseline			
Inhabitants by locality in 2007	515.290	486.391	-28.899
	(41.522)	(44.392)	
Household size 2007	3.681	3.331	-0.350^{***}
	(0.051)	(0.048)	
Index of quality of life 2007	87.400	91.529	4.129^{***}
	(0.151)	(0.216)	
Log of Average monthly expenditure 2007	13.652	14.086	0.434^{***}
	(0.533)	(0.372)	
Fixed controls			
Average socioeconomic classification (estrato)	2.162	3.352	1.190^{***}
	(0.135)	(0.119)	
Nearby transmilenio stations	2.667	6.034	3.367^{***}
	(0.445)	(0.552)	
Access to arterial roads	29.549	43.831	14.281^{***}
	(3.893)	(2.759)	
UPZs	51	59	110

Note: Significance: *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

I also evaluated whether there was indeed a tendency for hard discount stores, such as D1, Ara, and Justo & Bueno, to locate near OXXO establishments. In Appendix C, I show some of the results, and in general found that, for all years, these hard discount stores were significantly more prone to locate in UPZs treated with OXXO than in those without. With this evidence, I will include these chains as regressors mainly due to their probable influence on the dependent variables, as shown by Jaramillo López et al., 2021 in their investigation. However, I considered them in the regression as baseline covariates, meaning that I only took into account their presence in the UPZ in the quarter before the analysis in order to avoid confounding effects.

Although I considered hard discount chains as factors that may bias the estimation, other unobserved variables could also affect the results. For instance, public works can reduce pedestrian and vehicular traffic for certain periods within the analysis; a new park may attract different demographic flows; and the presence of other commercial establishments, such as bars, nightclubs, or grocery stores, may impact the estimation of the causal effect, especially if they open or close during the period of study. Nonetheless, some of these variables may remain constant from 2015 to 2022. Furthermore, other establishments may be controlled for by unit fixed effects, as they tend to be located in areas that maintain their core characteristics; for example, nightclubs often open in places already recognized as nightlife areas.

8 Empirical strategy

This research explores with two main econometric models the causal effect of OXXO openings on local crime activity in Bogotá. The first one I used is TWFE. This econometric strategy takes into account that the the arrival of OXXO's occurs in a staggered manner. Additionally, it inherently controls for fixed effects for unit (UPZ) and time. The former is useful for capturing time-invariant unobserved characteristics of each UPZ (such as geographic location, historical infrastructure or fixed socieconomic and urban characteristics during the period of analysis), while the latter absorbs common shocks that affect all units simultaneously (such as city-wide security policies or macroeconomic fluctuations). Although TWFE is and adequate method for estimating the average treatement effect (ATT) of OXXO on crime activity, two identification assumptions are needed to prevent bias. This ones will be considered later, but let us first introduce the basic regression form of TWFE:

$$Y_{i,t} = \alpha_i + \alpha_t + \beta^{twfe} D_{i,t} + \varepsilon_i \tag{2}$$

Where $Y_{i,t}$ represents the crime rate for UPZ i in quarter t. This rate corresponds to either a specific crime category or the aggregate of all crimes analyzed in this study. $D_{i,t}$ is a binary indicator that equals one if UPZ i has at least one OXXO store in quarter t , and zero otherwise. Furthermore, as Figure 5 suggests that intensity of treatment may be a factor that influences crime. Section 9 also evaluated $D_{i,t}$ as the number of OXXO establishments that an UPZ has at a certain quarter. Finally, α_i and α_t are the unit and time fixed effects respectively.

A critical identification assumption for the TWFE estimator is the homogeneity of treatment effects. This implies that these effects must be constant among treatment cohorts and over time (de Chaisemartin & D'Haultfœuille, 2023). This assumption may be violated if treatment effects vary based on UPZs' characteristics or if the impact of an OXXO opening on crime evolves as the store becomes established in the area. Consequently, to diagnose the potential bias arising from these heterogeneities, I employ the Goodman-Bacon decomposition (Goodman-Bacon, 2021). This theorem decompoeses the β^{twfe} as a weighted average of all possible 2×2 difference-in-differences comparisons among cohorts, and the primary concern in this decomposition is to avoid having negative weights as they indicate that the estimator may be biased from heterogeneous treatment effects (Goodman-Bacon, 2021).

Parallel trends is the other condition to guarantee an unbiased ATT, as it ensures that, in the absence of treatment, the average outcomes of the treated and control groups would have followed the same trajectory over time. To evaluate the validity of this assumption, I implement an Event Study design. This model requires no anticipation to treatment, and in this context it is highly unlikely that perpetrators or residents altered their behavior in expectation of an OXXO opening, as these commercial developments are typically not publicized far enough in advance to influence local criminal dynamics beforehand. The Events Study is estimated by using the following equation:

$$Y_{i,t} = \alpha_i + \alpha_t + \sum_{k=-K}^{-2} \gamma_k^{lead} D_{i,t}^k + \sum_{k=0}^{L} \gamma_k^{lag} D_{i,t}^k + \varepsilon_{i,t} \tag{3}$$

Equation(3) allows to analyze the significance of the coefficients γ_k^{lead} and suggests the existence of parallel trends before the arrival of OXXO. Additionally, the coefficients γ_k^{lag} reflect the temporal dynamics of the impact after the treatment.

Finally, for robustness purposes, I will also estimate the Callaway & Sant'Anna (CS) model. This approach accounts for treatment effect heterogeneity by estimating ATTs by cohort and period, subsequently combining these effects through a weighted average (Callaway & Sant'Anna, 2021; Rios-Avila et al., 2021). To achieve this, the model utilizes the following doubly robust estimator for the ATT of each cohort g at time t:

$$ATT(g,t) = \mathbb{E}\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)}}{\mathbb{E}\left[\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)}\right]}\right)(Y_t - Y_{g-1})\right] \tag{4}$$

Where t is a particular time period, G_g is a binary indicator for whether an observation belongs to cohort g, D_t is the treatment status at time t, and $(1 - D_t)$ identifies observations that have not yet been treated, serving as the control group. Additionally, the approach I chose uses Inverse Probability Weighting (IPW) to assign greater weight to control observations that more closely resemble the treated cohort G_g.

Subsequently, to aggregate all $ATT(g,t)$ estimates into a single summary parameter, the authors propose the following aggregation strategy:

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} w(g,t) \cdot ATT(g,t) \tag{5}$$

Where $w(g,t)$ represents the weights determined by the researcher. In this analysis, I employ a simple averaging scheme where $w(g,t)$ is the same for all treated periods.

The objective of this comparison is to determine if the Callaway & Sant'Anna (CS) estimates an Events Studies align with the Two-Way Fixed Effects (TWFE) results. A similar behavior between the two models would reinforce the validity of the TWFE estimates and suggest that the possible bias of heterogeneous effects is not a preoccupation (Callaway & Sant'Anna, 2021; Rios-Avila et al., 2021).

9 Results

Table 3: Effect of the opening of OXXO stores on crime rates

Crime rates	Theft to People	Sexual Offenses	Homicide	Theft to Motorbike	Theft to Vehicle
	(1)	(2)	(3)	(4)	(5)
Has OXXO	0.017	-0.011	-0.0206	-0.0022	-0.015
	(0.019)	(0.014)	(0.013)	(0.014)	(0.015)
Constant	0.301***	0.006	0.012	0.02	0.018
	(0.008)	(0.010)	(0.022)	(0.016)	(0.017)
Chain Stores	no	no	no	no	no
Observations	3,584	3,584	3,584	3,584	3,584
R-squared	0.000	0.222	0.375	0.460	0.480
Mean never treated	0.259	0.259	0.259	0.259	0.259

Note: Standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area.

10 Robustness

spill over effects buffers for the spill over effects, etc. treat controls and treated groups as buffers

11 Discussion

12 Conclusions

Table 4: Events Studies of TWFE and CS on different types of crime rates

Crime rates	Theft People		Sexual Offenses		Homicide		Theft Motorbike		Theft Vehicle	
Method	TWFE	CS	TWFE	CS	TWFE	CS	TWFE	CS	TWFE	CS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Year after treatment = -3	-0.0216	...	0.0330	...	0.006	...	0.040	...	...	...
	(0.0232)		(0.0334)		(0.024)		(0.016)			
Year after treatment = -2	0.0195	...	0.0459	...	-0.012	...	0.007	...	...	...
	(0.0229)		(0.0267)		(0.021)		(0.020)			
Year after treatment = 0	0.0103	...	0.0215	...	-0.013	...	0.006	...	...	...
	(0.0166)		(0.0180)		(0.017)		(0.027)			
Year after treatment = 1	-0.0130	...	-0.0252	...	-0.029	...	0.003	...	...	...
	(0.0135)		(0.0192)		(0.023)		(0.023)			
Year after treatment = 2	-0.0356*	...	-0.0605*	...	-0.071**	...	-0.034	...	...	...
	(0.0261)		(0.0299)		(0.033)		(0.020)			
Year after treatment = 3	-0.0690**	...	-0.118***	...	–	...	–	...	...	...
	(0.0273)		(0.0409)		–		–			
ATT	-0.0206	...	-0.0022	...	-0.0214	...	0.0014	...	...	...
SE ATT	(0.0193)	...	(0.0195)	...	(0.019)	...	(0.018)	...	...	...
Observations	684	...	684	...	3,520	...	3,520	...	...	...

Note: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. ATT is the average effect of the treatment on the treated. For TWFE, this is the coefficient presented in Table 3 and for CS is the weighted average of the ATTs by cohort.

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13 Appendix

A Detailed description of variables

Table A.1: Data sources

Variable	**Description**	**Data source**
Dependent Variable		
Aggregated Crime rate	Crime rate per 10,0000 inhabitants	SIEDCO
Theft Crime rate	Theft rate per 10,0000 inhabitants	SIEDCO
Theft Crime rate	Theft rate per 10,0000 inhabitants	SIEDCO
Homicide rate	Homicide rate per 10,0000 inhabitants	SIEDCO
Sexual violence crime rate	Sexual violence crime rate per 10,0000 inhabitants	SIEDCO
Theft of Motorcycles crime rate	Theft of Motorcycles crime rate per 10,0000 inhabitants	SIEDCO
Theft of Vehicles crime rate	Theft of Vehicles crime rate per 10,0000 inhabitants	SIEDCO
Staggered Treatment Variable		
Has OXXO (= 1)	If the UPZ has OXXO(s) that quarter	RUES and Google Maps API
Staggered Controls of Hard Discount Stores		
Has D1s (=1)	If the UPZ has D1(s) in the quarter before the quarter analyzed	RUES y Google Maps API
Has ARAs (=1)	If the UPZ has ARA(s) in the quarter before the quarter analyzed	RUES y Google Maps API
Baseline Controls		

Table A.1: Data sources

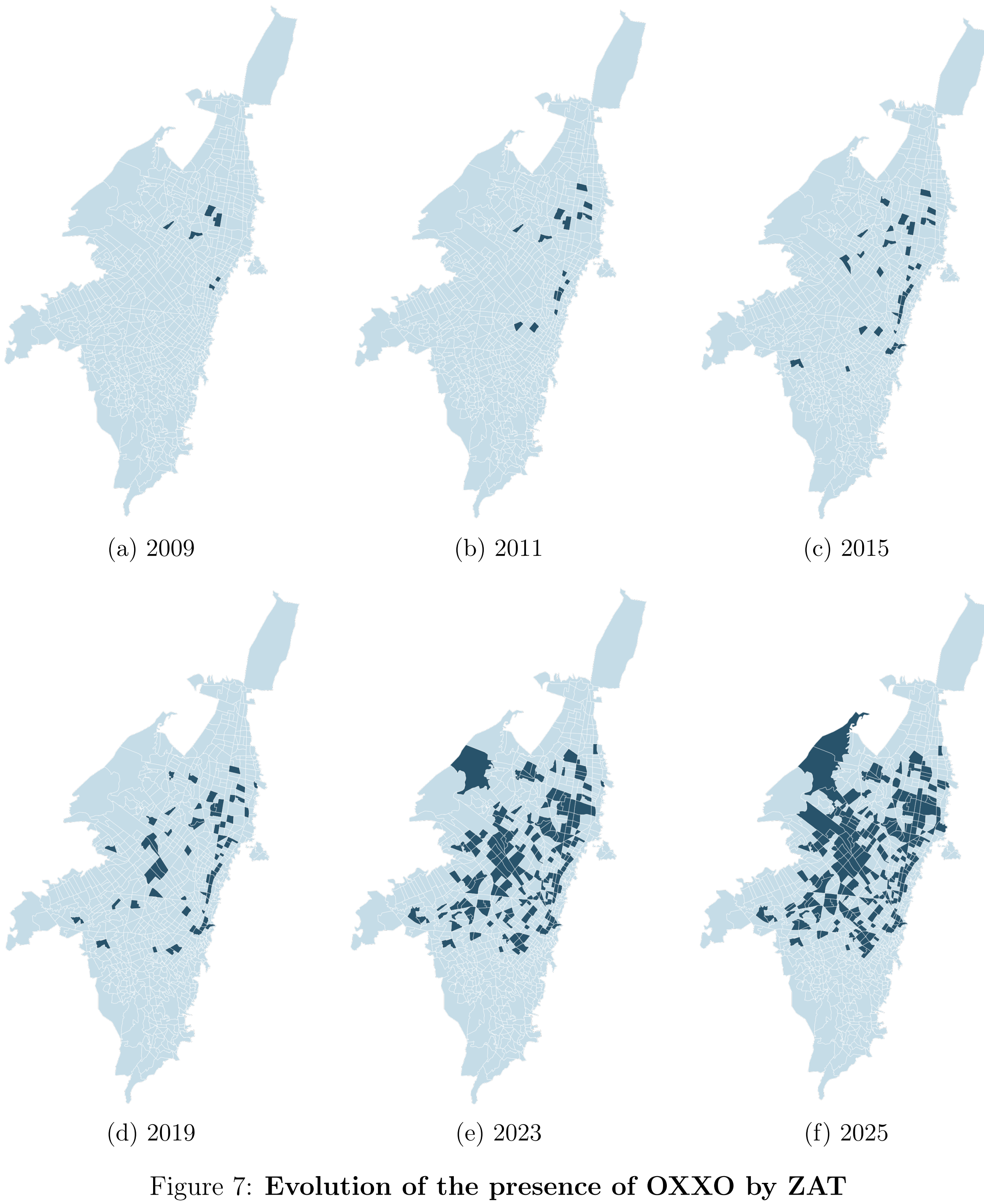
Variable	Description	Data source
Inhabitatnsper locality in 2007	Número de habitantes por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Cantidad de personas por hogar 2007	Número promedio de personas en el hogar por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Indice de condiciones de vida 2007	Indice de condiciones de vida promedio por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Gasto promedio mensual 2007	Gasto promedio mensual en el hogar por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Controles fijos en el tiempo		
Estrato promedio del ZAT	Estrato promedio del ZAT, que oscila de 1 a 6	Mapas Bogotá - Infraestructura de Datos Espaciales de Bogotá (IDECA)
Estaciones de transmilenio cercanas	Cantidad de estaciones de transmilenio dentro o cercanos al ZAT	Mapas Bogotá - Infraestructura de Datos Espaciales de Bogotá (IDECA)
Cantidad acceso a vías arteriales	Cantidad de vías arteriales dentro o cercanos al ZAT	Mapas Bogotá - Infraestructura de Datos Espaciales de Bogotá (IDECA)
Control spillover		

Table A.1: Data sources

Variable	**Description**	**Data source**
Cantidad de Oxxos cercanos fuera del ZAT	Cantidad de OXXOs fuera del ZAT, que se encuentran cerca de este.	RUES y Google Maps API

Nota: No existe controles de línea base por ZAT, por lo que solo se pudo obtener índices previos al 2011 por localidad (ECDV 2007). El control "spillover" tiene en cuenta que hay ZATs que no tienen OXXOs, pero presentan OXXOs cercanos que pueden afectar la dinámica económica y laboral de la zona analizada. Finalmente, cuando de habla de OXXOs, vías arteriales y estaciones de transmilenio cercanos, significa que se encuentran a menos de 400m del ZAT.

B Evolution of OXXO stores in Bogotá divided by Transport Analysis Zones (ZATs)



(a) 2009 (b) 2011 (c) 2015

(d) 2019 (e) 2023 (f) 2025

Figure 7: **Evolution of the presence of OXXO by ZAT**

C Staggered controls of discount store chains

Table C.1: Means differences of hard discount store staggered controls for the 2018-4 cohort

	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.477 (0.054)	0.477 (0.079)	0.172***
Quantity of D1s	1.233 (0.214)	3.250 (0.543)	2.017***
Has Aras	0.256 (0.047)	0.714 (0.087)	0.458***
Quantity of Aras	0.384 (0.089)	1.429 (0.264)	0.216***
Has J&B	0.895 (0.050)	1.964 (0.096)	1.069
Quantity of J&B	0.062 (0.256)	0.278 (1.202)	0.216***
UPZs	86	28	114

Nota: Significancia: *** p<0.01, ** p<0.05, * p<0.1.

Table C.2: Means differences of hard discount store staggered controls for the 2022-4 cohort

	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.491 (0.067)	0.881 (0.042)	0.390***
Quantity of D1s	1.053 (0.191)	4.458 (0.512)	3.405***
Has Aras	0.298 (0.061)	0.627 (0.063)	0.329***
Quantity of Aras	0.386 (0.089)	1.712 (0.287)	1.326***
UPZs	57	59	116

Nota: Significancia: *** p<0.01, ** p<0.05, * p<0.1.

D Distribution of other crimes in Bogotá over time